

UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET
DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE XII

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG

ABRIL/2026

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Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Cálculo Integral e Diferencial do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

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Atividade 12.

1-c. $\operatorname{tg} x = \frac{\operatorname{sen} x}{\cos x}$

$$\cos x \cdot \cos x = \operatorname{sen} x \cdot (-\operatorname{sen} x) = -\operatorname{sen}^2 x = -\frac{\operatorname{sen}^2 x}{\cos^2 x} = \frac{-\operatorname{sen}^2 x}{\cos^2 x} = \frac{-1 + \cos^2 x}{\cos^2 x}$$

d. $\operatorname{sec} x = \frac{1}{\cos x}$

$$0 \cdot \cos x = 1 \cdot (-\operatorname{sen} x) = -\frac{\operatorname{sen} x}{\cos^2 x} = \frac{-\operatorname{sen} x}{\cos^2 x} = \frac{-\operatorname{sen} x}{\cos^2 x} = \frac{-\operatorname{sen} x}{\cos^2 x} = \frac{-\operatorname{sen} x}{\cos^2 x}$$

e. $\operatorname{cotg} x = \frac{\cos x}{\operatorname{sen} x}$

$$-\operatorname{sen} x \operatorname{sen} x = \cos x \cos x = \cos^2 x = -\operatorname{sen}^2 x - \cos^2 x = -(\operatorname{sen}^2 x + \cos^2 x) = \frac{-1}{\operatorname{sen}^2 x} = -\operatorname{cosec}^2 x$$

f. $\operatorname{cosec} x = \frac{1}{\operatorname{sen} x}$

$$0 \cdot \operatorname{sen} x = 1 \cdot \cos x = \frac{\cos x}{\operatorname{sen}^2 x} = -\operatorname{cosec} x \cdot \operatorname{cotg} x$$

2. $f(x) = 2^x$
 $f'(x) = 2^x \ln 2$

$f(x) = 5^x$
 $f'(x) = 5^x \ln 5$

$f(x) = \pi^x$
 $f'(x) = \pi^x \cdot \ln \pi$

$f(x) = e^x$
 $f'(x) = e^x$

6. $g(x) = \log_3 x$
 $g'(x) = \frac{1}{x \ln 3}$

$g(x) = \log_5 x$
 $g'(x) = \frac{1}{x \ln 5}$

$g(x) = \log_{\pi} x$
 $g'(x) = \frac{1}{x \ln \pi}$

$g(x) = \ln x$
 $g'(x) = \frac{1}{x}$

3. $f(x) = \operatorname{sen} x$
 $x = 0$

$f'(0) = \operatorname{sen} 0 = 0$
 $P(0,0)$

$f'(x) = \cos x$
 $f'(0) = \cos 0 = 1$

$y - 0 = 1(x - 0)$
 $y = x$

5. $f(x) = \operatorname{tg} x$ $f(0) = \operatorname{tg}(0) = 0$
 $x = 0$

$f'(x) = \operatorname{sec}^2 x$
 $m = f'(0) = \operatorname{sec}^2(0) = \frac{1}{\cos^2(0)} = 1$

$y - 0 = 1(x - 0)$
 $y = x$

4.9

$$\frac{x + \sin x}{x - \cos x}$$

$$\frac{(1 + \cos x) \cdot (x - \cos x) - (x + \sin x) \cdot (1 + \sin x)}{(x - \cos x)^2} =$$

$$\frac{x - \cos x + x \cos x - \cos^2 x - x - x \sin x - \sin x - \sin^2 x}{(x - \cos x)^2} =$$

$$\frac{-\cos x - \sin x + x \cos x - x \sin x - \cos^2 x - \sin^2 x}{(x - \cos x)^2} =$$

$$\frac{x \cos x - x \sin x - \cos x - \sin x - 1}{(x - \cos x)^2}$$

$$\frac{x(\cos x - \sin x) - (\cos x + \sin x) - 1}{(x - \cos x)^2}$$

14.

a. $x e^x \cos x$

$f(1) = x$

$f(2) = e^x$

$f(3) = \cos x$

$$f'(x) = 1 \cdot e^x \cos x + x e^x \cos x + x e^x - \sin x$$

$$= e^x (\cos x + x \cos x + x - \sin x)$$

b. $(x^2 \cos x)(1 + \ln x)$

$f(1) = x^2$

$f(2) = \cos x$

$f(3) = 1 + \ln x$

$$f'(x) = (2x) \cdot (\cos x) \cdot (1 + \ln x) + x^2 \cdot (-\sin x) \cdot (1 + \ln x) + x^2 \cos x \cdot \frac{1}{x}$$

$$f'(x) = 2x \cos x (1 + \ln x) - x^2 \sin x (1 + \ln x) + x \cos x$$

c. $e^x \sin x \cos x$

$$f'(x) = e^x \sin x \cos x + e^x \cos x \sin x + e^x \cos x - \sin x$$

$$= e^x (\sin x \cos x + \cos^2 x - \cos x \sin x)$$

$f(1) =$

$f(2) =$

$f(3) =$

d. $(1 + \sqrt{x}) e^x \tan x$ $f'(x) = \frac{1}{2\sqrt{x}} e^x \tan x + (1 + \sqrt{x}) e^x \tan x + (1 + \sqrt{x}) e^x \sec^2 x$

$$= e^x \left(\frac{1}{2\sqrt{x}} \tan x + (1 + \sqrt{x}) \tan x + (1 + \sqrt{x}) \sec^2 x \right)$$

5. a. $y = u+1$
 $\ln u$

$$y' = 1 \cdot \ln u - (u+1) \cdot \frac{1}{u}$$

$$y' = \ln u - u+1 \cdot \frac{1}{u} = \ln u - 1 + \frac{1}{u}$$

b. $a = e^t \ln t$

$$a' = e^t \ln t + e^t \cdot \frac{1}{t}$$

$$e^t (\ln t + \frac{1}{t})$$

c. $y = \sqrt[3]{u} \sec u$

$$y' = \frac{1}{3\sqrt[3]{u^2}} \sec u + \sqrt[3]{u} \sec u \tan u$$

$$\sec u \left(\frac{1}{3\sqrt[3]{u^2}} + \sqrt[3]{u} \tan u \right)$$

d. $x = e^t \cos t$

$$x' = e^t \cos t + e^t (-\sin t)$$

$$x' = e^t (\cos t - \sin t)$$

4. $f(x) = xt^3$

$$\frac{d}{dt} (xt^3) = x \cdot 3t^2 + t^3 \frac{dx}{dt}$$

$$3 \cdot 2^3 + 1 \cdot 3(2)^2$$

$$\frac{dy}{dt} \Big|_{t=2} = 36$$

12. $y = e^x \cos x$

$$\frac{dy}{dx} = e^x \cos x - e^x \sin x$$

$$\frac{d^2y}{dx^2} = (e^x \cos x - e^x \sin x) - (e^x \sin x + e^x \cos x)$$

$$e^x \cos x - e^x \sin x - e^x \sin x - e^x \cos x = 0$$

$$16 - a. \frac{d}{dt} \left(t^2 \frac{dx}{dt} \right) = 2t \frac{dx}{dt} + t^2 \frac{d^2x}{dt^2}$$

$$\frac{d}{dt} \left(t^2 \frac{dx}{dt} \right) = 2t \frac{dx}{dt} + t^2 \frac{d^2x}{dt^2}$$

Regra do produto

$$b. \frac{d}{dt} \left(x \frac{dx}{dt} \right) = \frac{dx}{dt} \frac{dx}{dt} + x \frac{d^2x}{dt^2}$$

$$u = x \quad \left(\frac{dx}{dt} \right)^2 + x \frac{d^2x}{dt^2}$$

$$v = \frac{dx}{dt}$$

$$\frac{du}{dt} = \frac{dx}{dt}$$

$$\frac{dv}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2x}{dt^2}$$